

# Japanese speakers can infer specific sub-lexicons using phonotactic cues



Shigeto Kawahara & Gakuji Kumagai

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The PDF of this handout is available from the QR code above.

The written-up version is available at <https://ling.auf.net/lingbuzz/008145>

# Background

- Phonotactic restrictions do not usually hold uniformly across the entire lexicon.
- A famous example: Japanese has several “lexical strata”, which are sensitive to phonotactic restrictions to different degrees (Ito & Mester 1995, 1999). E.g. voiced obstruent geminates are not allowed, except in loanwords.
- A question that is addressed in some recent studies: how fined-grained can these sub-lexicons be? (e.g. Gouskova et al. 2015)

# Two broad hypotheses

- The “classic view”: Languages only allow “general” divisions (perhaps with independent motivations, historical and/or orthographic).
  - Japanese: native, Sino-Japanese, recent loanwords, mimetics (Ito & Mester 1999).
  - English: Germanic vs. Latinate (Chomsky & Halle 1968).
- The emerging view: Sublexicons can be as specific as “a set of roots to which a particular affix is attached to” (Gouskova et al. 2015).

# The sublexal phonology/co-phonology

- “learning lexically specific morphological and phonological rules involves separating the lexicon into **sublexicons**. Phonological generalizations about the application of such rules are encoded in part as phonotactic grammars learned over sublexicons.” (Gouskova & Ahn to appear, page 6).
- “Cophonology Theory is motivated by the non-uniformity of the phonological grammar of single languages. In this theory, every language contains multiple phonological sub-grammars which apply in different morphosyntactic environments. To date, cophonologies have been indexed to **part of speech** (Anttila 2002; Smith 2011) and to **specific morphological constructions** (Orgun 1996; Inkelas 1998; Inkelas and Zoll 2005, 2007)” (Sande 2020). [**EMPHASIS OURS**].

# Our hunch

- Some sound sequences in Japanese are over-represented -- or restricted to -- a very specific word class (i.e. a sub-lexicon).
- Geminate [rr]s are generally not allowed in Japanese, but they often appear on “Italian restaurant menus”.
- Geminate [hh]s (surrounded by identical vowels) are also very rare, except in Dutch/German names.
- Snack names often contain singleton [p]s.

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## トルテッリ (tortelli)

ラビオリに対して、1枚の生地で閉じているのがトルテッリです。



## カンネッローニ (cannelloni)

詰め物を筒状に巻いたパスタの種類です。ほうれん草を包んだりチキンを包んだりします。トマトソースをかけます。日本ではあまり見かけないユニークなパスタです。誰かに振る舞ったら驚かれるかもしれません。



タリアテッレ

# Expanding on the last point



- In July 2023, I [Shigeto] published this book, “Why snack names contain a lot of [p]s?”.
- This is based on an actual lecture that I gave to elementary school students.
- Since I had never taught linguistics to elementary school students, prior to the event, I had asked the participants to send me “whatever questions they have about languages”.

# Expanding on the last point



- One of the questions was, “I feel like [p] is often used in snack names, like *papiko* and *poiuru*. Why is that?”
- I thought that this was a fantastic question. And she provided many actual examples!
- So I provided some possible explanations (not directly relevant here).
- The publisher thought that this is the catchiest question, and it became the main title of the book.

よくおかしの商品名に「パ」とか「ピ」が付けられているけどなんで?  
川原 じゃあ、次の質問。「よくお菓子の商品名にパとかピが付けられているけどなんで?」。  
わこ、素晴らしい質問だね。わこがあげてくれた例は「パピコ」「パイの実」「チョコパイ」「アボロ」「ポイフル」ですね。

# Expanding on the last point



- But a few months later after the publication of this book, I realized that “why X” **presupposes** that “X is correct”.
- But I did not actually test whether X is correct or not in that very book.
- As a linguist, I decided that it is my responsibility to test X, the presupposition.
- And this motivation converged with the theoretical question discussed at the outset.

# Three hunches tested



1. Singleton [p]s are overrepresented in snack names.
2. Geminate [rr]s almost exclusively appear in Italian loanwords.
3. Geminate [hh]s appear often in German names, but not anywhere else.



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# Experiment 1

- Japanese speakers were presented with a target word containing the phonotactic cue (e.g. [pariko]) and a control word not containing that phonotactic cue (e.g. [tariko]) .
- They were asked to choose whether each name is suitable for the target category (e.g. a European snack) or a control category (e.g. a cosmetic brand).
- We presented one item per trial, not in a pair (i.e. a forced choice task, not a 2AFC).

|              | Target    | Control   |
|--------------|-----------|-----------|
| [p]=snack    | [pariko]  | [tariko]  |
|              | [pasomi]  | [tasomi]  |
|              | [penaro]  | [tenaro]  |
|              | [posine]  | [tosone]  |
|              | [pamore]  | [kamore]  |
|              | [parase]  | [karase]  |
|              | [pesemo]  | [tesemo]  |
|              | [ponoru]  | [koniru]  |
| [rr]=Italian | [metorra] | [metobba] |
|              | [makorro] | [makobbo] |
|              | [nesorra] | [nesodda] |
|              | [nokirre] | [nokidde] |
|              | [temerro] | [temeddo] |
|              | [tamirra] | [tamigga] |
|              | [kanorri] | [kanoggi] |
|              | [tonorre] | [tonogge] |
| [hh]=German  | [bohho]   | [boppo]   |
|              | [kuϕϕu]   | [kuppu]   |
|              | [gehhe]   | [geppe]   |
|              | [gahha]   | [gatta]   |
|              | [rehhe]   | [rette]   |
|              | [biççi]   | [bikki]   |
|              | [nahha]   | [nakka]   |

## The stimuli

- Target=[p]-initial words; control=[t/k]-initial words.
- Target items contain [rr]; control items contain a voiced obstruent geminate.
- Target items contain [hh], surrounded by identical vowels; control items contain a voiceless obstruent geminate.

|              | Target    | Control   |
|--------------|-----------|-----------|
| [p]=snack    | [pariko]  | [tariko]  |
|              | [pasomi]  | [tasomi]  |
|              | [penaro]  | [tenaro]  |
|              | [posine]  | [tosone]  |
|              | [pamore]  | [kamore]  |
|              | [parase]  | [karase]  |
|              | [pesemo]  | [tesemo]  |
|              | [ponoru]  | [koniru]  |
| [rr]=Italian | [metorra] | [metobba] |
|              | [makorro] | [makobbo] |
|              | [nesorra] | [nesodda] |
|              | [nokirre] | [nokidde] |
|              | [temerro] | [temeddo] |
|              | [tamirra] | [tamigga] |
|              | [kanorri] | [kanoggi] |
|              | [tonorre] | [tonogge] |
| [hh]=German  | [bohho]   | [boppo]   |
|              | [kuϕϕu]   | [kuppu]   |
|              | [gehhe]   | [geppe]   |
|              | [gahha]   | [gatta]   |
|              | [rehhe]   | [rette]   |
|              | [biççi]   | [bikki]   |
|              | [nahha]   | [nakka]   |

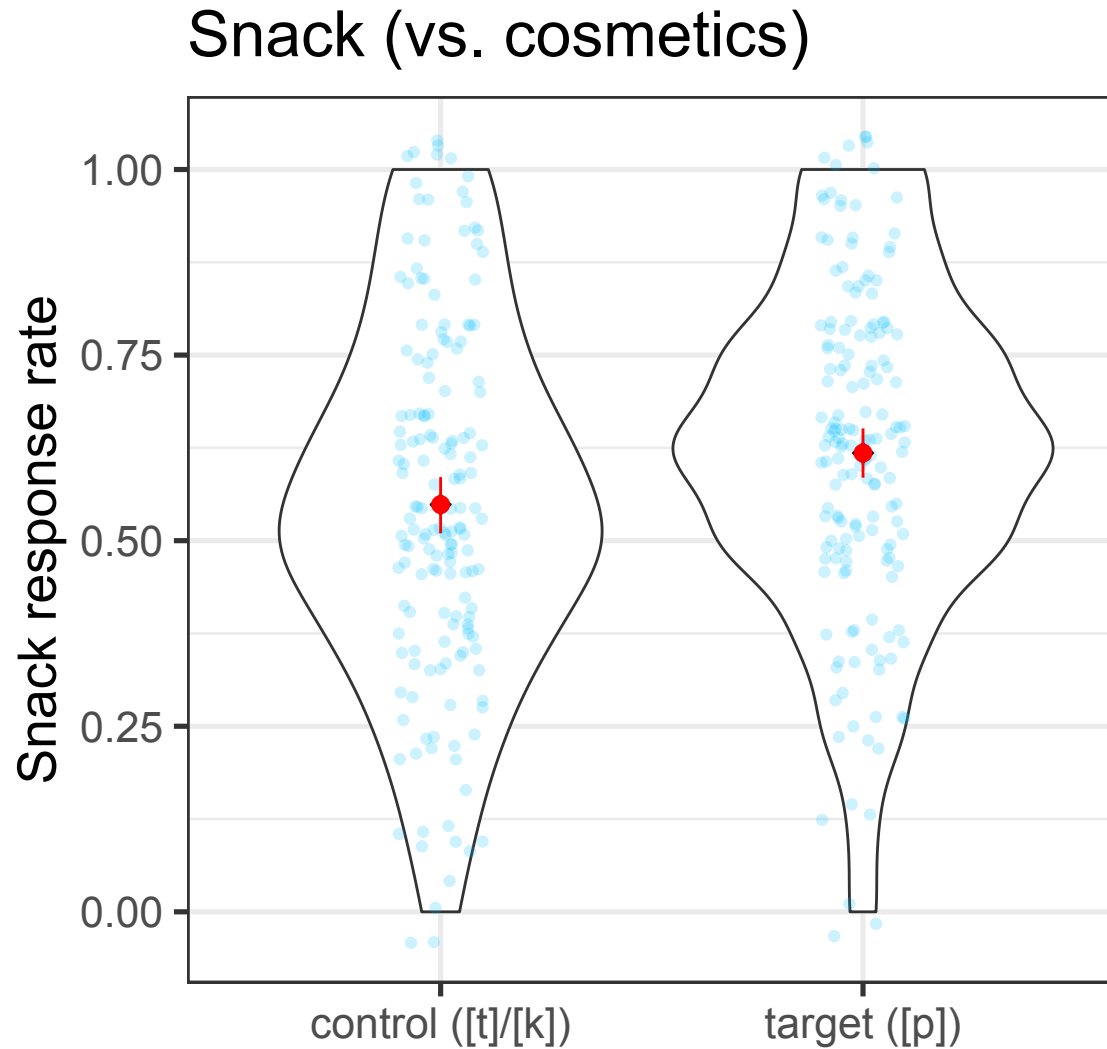
## The task

- Snack names or cosmetic names?  
(they are both written in the *katakana* orthography)
- Italian restaurant names or French restaurant names?
- German names or English names?

# Experiment details

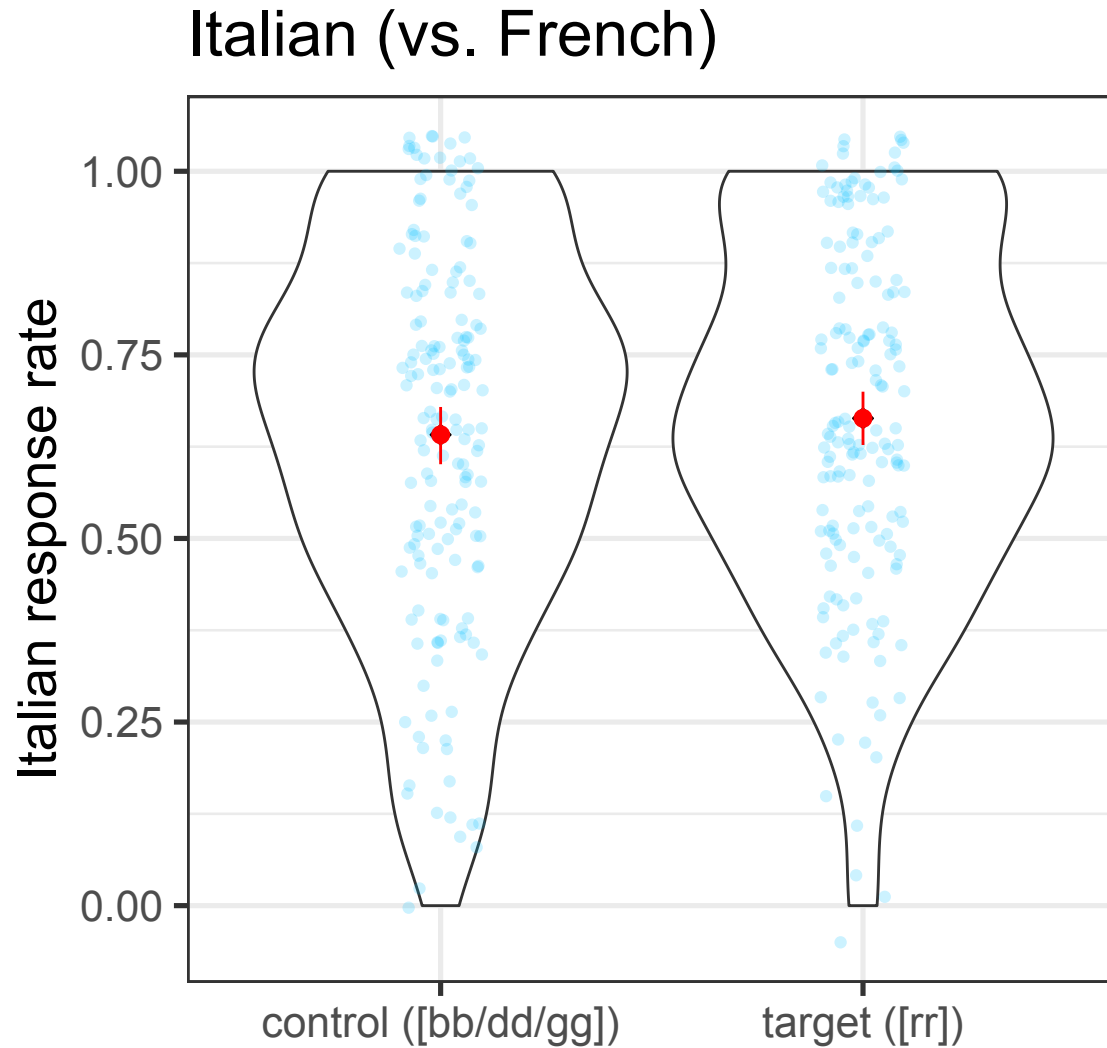
- Data from 162 participants were collected.
- Administered online using SurveyMonkey.
- The stimuli were presented in the *katakana* orthography.
- The data was analyzed using a Bayesian mixed effects logistic regression model. Full details are available on an osf repository
- [https://osf.io/97zc5/?view\\_only=4efac77f1b1a4e7c80798f7efbe86160](https://osf.io/97zc5/?view_only=4efac77f1b1a4e7c80798f7efbe86160)
- We present the posterior probability of the slope coefficient ( $\beta_1$ ) to be positive.

# Results



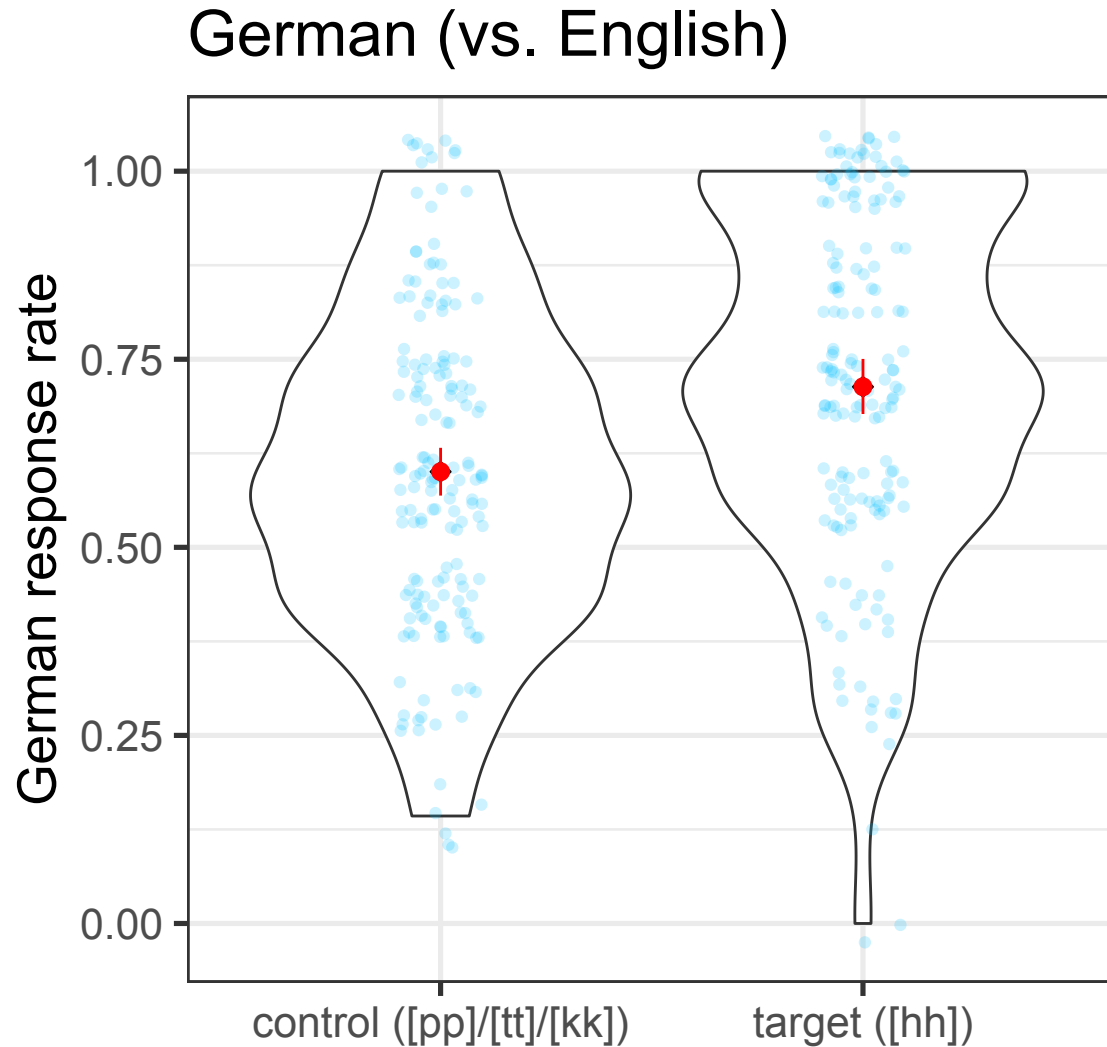
- Those items that contain [p] were more likely judged to be snack names than the control items.
- $p(\beta_1 > 0) = 0.95$ .

# Results



- Those items that contain [rr] were more likely judged to be Italian names than the control items, but not by much.
- $p(\beta_1 > 0) = 0.82$ .
- So not as clear as the association between [p] and snacks.

# Results



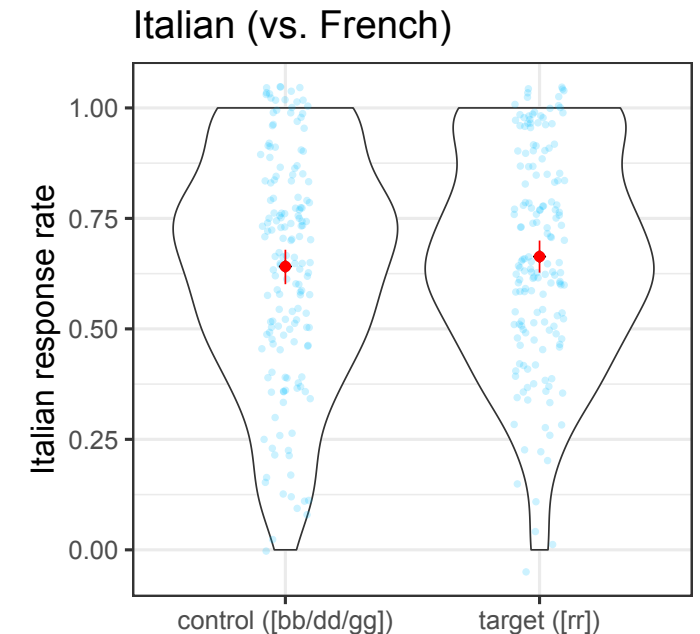
- Those items that contain [hh] were more likely judged to be German names than the control items.
- $p(\beta_1 > 0) = 0.95$ .

# Discussion

- Japanese speakers can associate nonce words with a particular phonotactic cue to a very specific lexical class.
- The connection between [rr] and Italian restaurant names was not very robust, however.

# Experiment 2

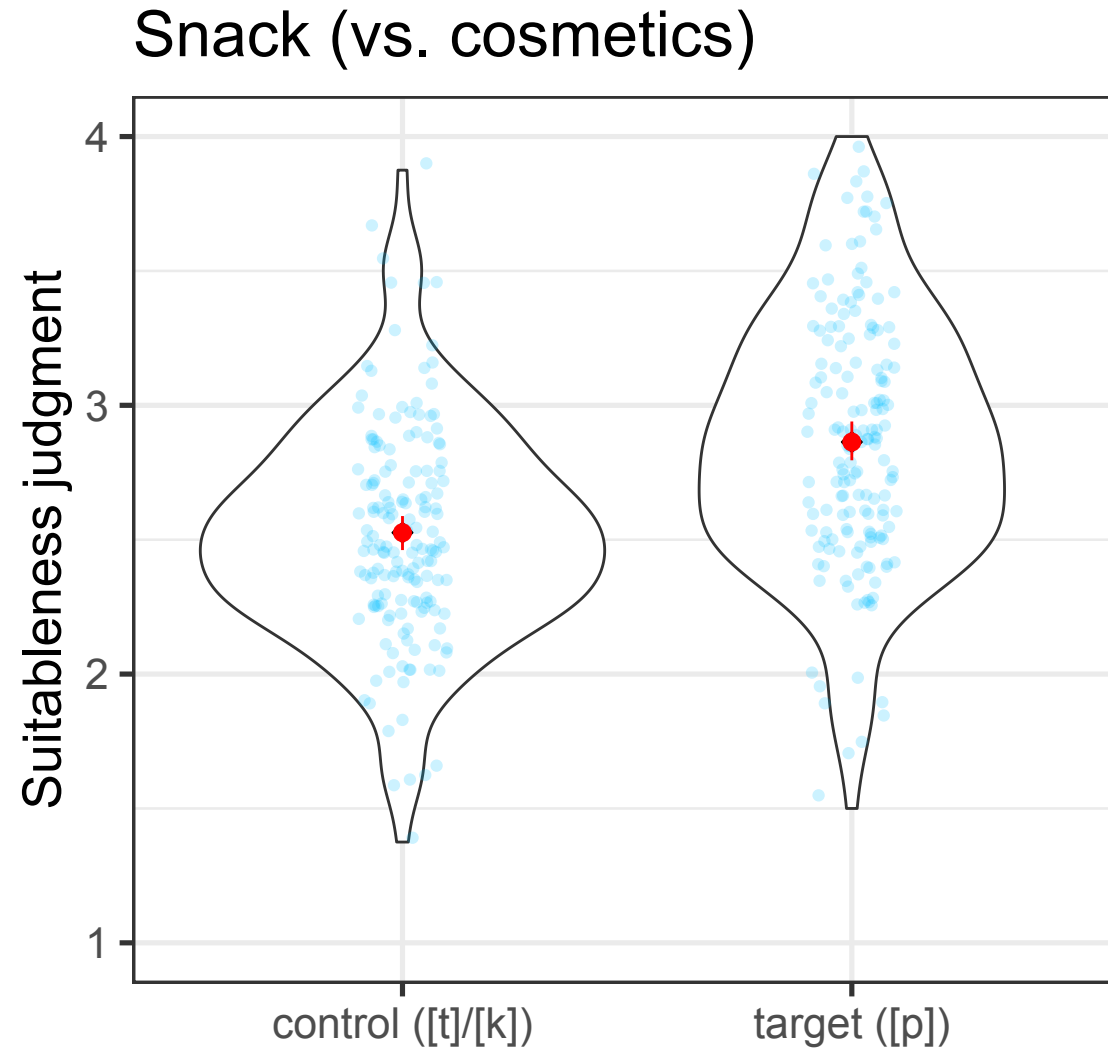
- Question: the association between [rr] and Italian names was not quite as robust as the other two effects.
- Control items (containing voiced obstruent geminates) were associated with Italian names with the probability that is much higher than chance.
- Maybe Japanese speakers do not want to use those names with voiced obstruent geminates for French restaurant names.



# Experiment 2

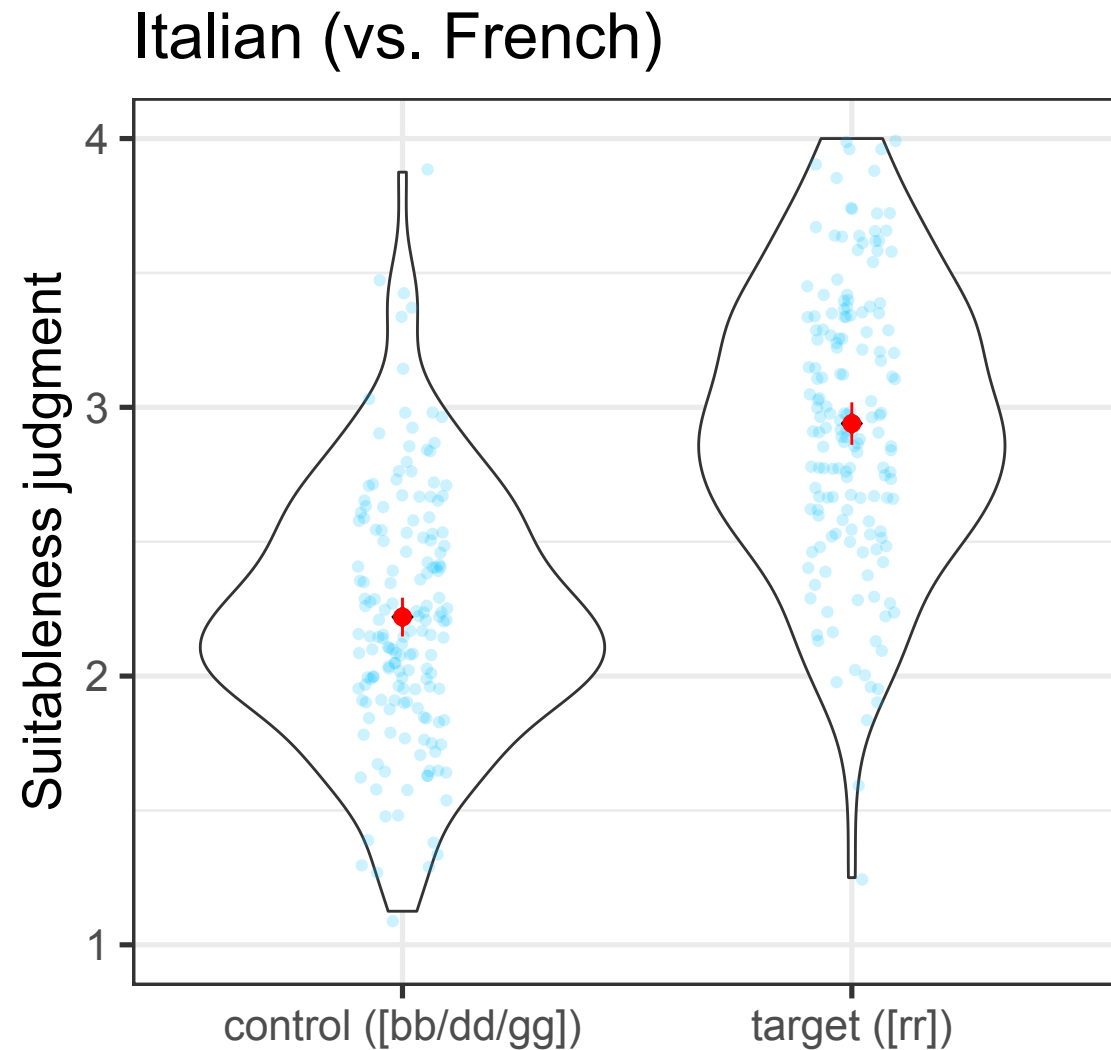
- This time, we presented each stimulus and asked how suitable each name is for the three respective target categories on a 4-point scale.
- E.g. How suitable is [pariko] as a snack name? How suitable is [bohho] as a German person name?
- 162 native speakers of Japanese.
- Experiment 1 contained [geppe] as one of the control items, but we excluded it because it may have reminded the Japanese speakers of [gebberusu], a (in)famous historical German figure.
- We used an *ordinal* logistic regression this time (Bayesian, mixed effects).

# Results



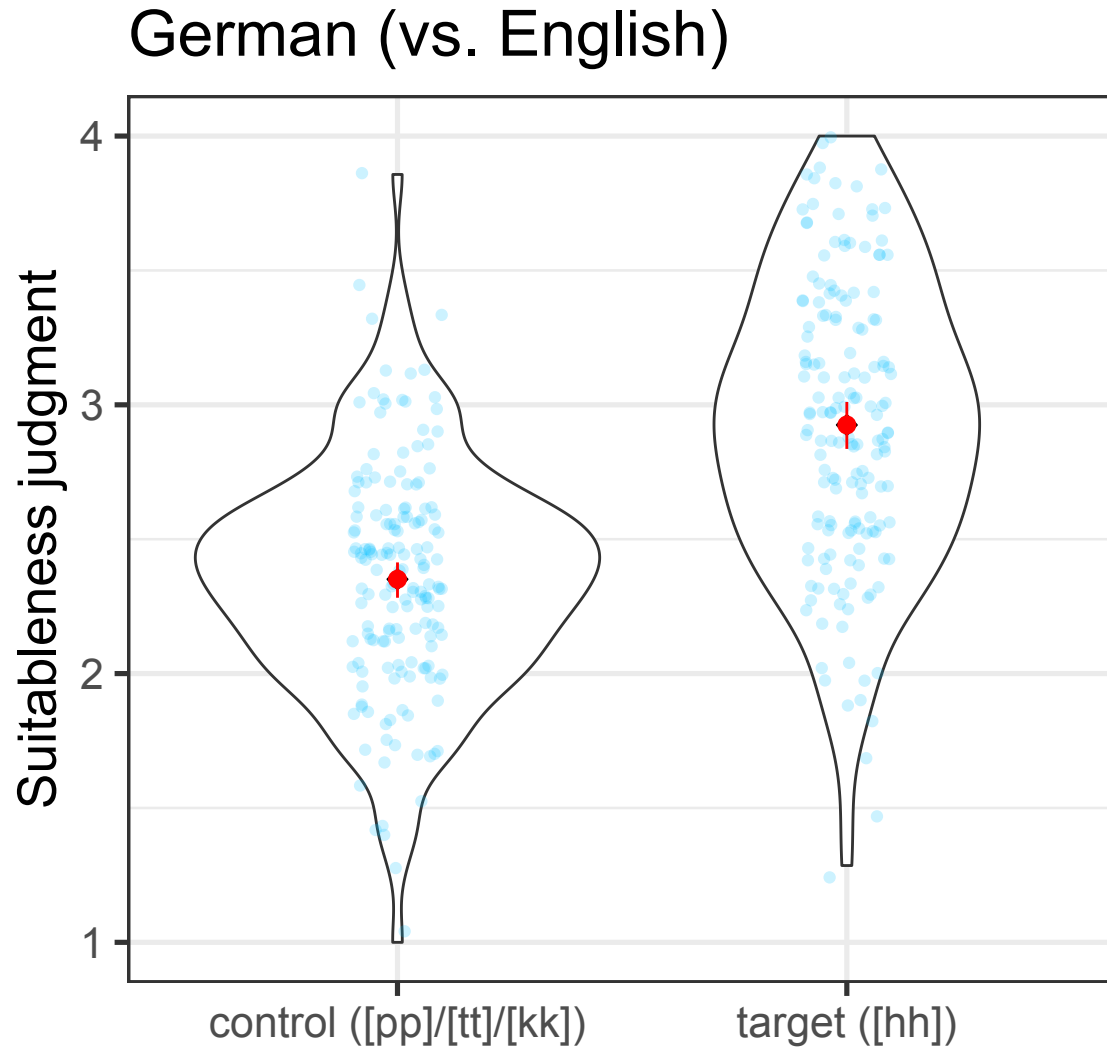
- Those items that contain [p] were judged to be more suitable for snack names than the control items.
- $p(\beta_1 > 0) = 0.91$ .

# Results



- Those items that contain [rr] were judged to be more suitable for Italian names than the control items.
- $p(\beta_1 > 0) = 1$ .

# Results



- Those items that contain [hh] were more judged to be suitable for German names than the control items.
- $p(\beta_1 > 0) = 1$ .

# Conclusions

- Japanese speakers can infer, based a particular phonotactic cue, a very specific subpart of the lexicon, that is something as specific as “snack names” or “Italian names” or “German names”.
- In this sense, the traditional lexical stratification into native, SJ, loanwords and mimetics does not suffice.
- This is compatible with the view put forward by Gouskova et al. (2015) and others.
- Any other cues we can test? [ttsa] limited to Italian names?

# What does this mean theory-wise?

- Express(X) (Alderete & Kochetov 2017): Use a particular type of sound to express a particular type of meaning.
- Express(PAL <for small>): Use palatal consonants or [i] -- sounds with high F2 -- to express smallness.
- Express ([p] <for snack>), Express([rr] <for Italian>), Express([hh] <for German>).

| /tariaterre/ (Italian) | Express([rr] <for Italian>) | *[rr] | Faith |
|------------------------|-----------------------------|-------|-------|
| => [tariaterre]        |                             | *     |       |
| [tariatere]            | *!                          |       | *     |

| /tariaterre/  | Express([rr] <for Italian>) | *[rr] | Faith |
|---------------|-----------------------------|-------|-------|
| [tariaterre]  |                             | *!    |       |
| =>[tariatere] |                             |       | *     |

- Express(X) constraints are formulated as markedness constraints, reflecting the intuition that “we should use sound X to sound like Y”.
- But we can formulate the same constraint as a faithfulness constraint.

And...

- I (Shigeto) was **not** wrong about the “[p]=snacks” association.
- With hindsight, however, I shouldn’t have allowed the publisher to use the title “Why X”, without examining the truth value of “X”.



- But at least I am being honest here. I admit that I made a mistake and examined my “hidden” assumption.
- It is amazing how often publishers use the “Why X”/“How-X” titles, presupposing that X is truth-conditionally correct. As a linguist, we should probably make a public warning about this practice.

2024.02.22 # エッセイ

## 言語学者と考える、音とことばの秘密。お菓子の名前は本当に「パピプペポ」が似合うのか？



川原 繁人

慶應義塾大学言語文化研究所教授  
言語学者

プロフィール

「お菓子の名前には「パピプペポ」がつくことが多いのはなんで？」1人の小学生の疑問から深まっていく、音とことばの秘密。『群像』2024年3月号より、言語学者・音声学者の川原繁人さんのエッセイ「お菓子の名前は本当に「パピプペポ」が似合うのか？」をお届けします。

### アクセスランキング

- 1 「源氏物語」はなぜ世界で読まれる名作になったのか？レディ・ムラサキ＝紫式部の時空を超え…
- 2 言語学者と考える、音とことばの秘密。お菓子の名前は本当に「パピプペポ」が似合うのか？
- 3 「古事記」はこんなに面白かったのか！町田康の超絶語訳で描く人間臭い神々の魅力
- 4 なぜ「事務的ミスの積み重ね」と述べれば、裏金が許されるのか？話題書『事務に踊る人々』を…
- 5 世界が絶賛した英訳「源氏物語」が現代日本に転生するまで…『レディ・ムラサキのティーバー…

Thank you